

Relations

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Overview

- 1 Sequences (4.2)
- 2 Binary Relations (4.4)
- 3 A side note: defining sequences

Definition

Definition: A *sequence* is an ordered list of items.

Like a set: Collection of items.

Unlike a set: Order matters, can have repeats.

Simple example: Cartesian coordinates like (x, y) .

Definition: The *empty sequence* is a sequence of length 0 and is denoted λ .

Operation on sets: Product

Definition: The product of two sets A and B is written $A \times B$ and is the set of all length-two sequences where the first element of the sequence comes from A and second from B .

Example:

■ $X = \{1, 2\}$

■ $Y = \{0, 2, 4\}$

■ $X \times Y = \{(1, 0), (1, 2), (1, 4), (2, 0), (2, 2), (2, 4)\}$

■ $|X \times Y| = 6 = |X| \times |Y|$

Extends to chains of products: $S_1 \times S_2 \times S_3 \times \cdots S_k$.

If $S_1 = S_2 = S_3 = \cdots = S_k = S$, can write simply S^k .

Example: $\{a, b\}^3 = \{(a, a, a), (a, a, b), (a, b, a), (a, b, b), (b, a, a), (b, a, b), (b, b, a), (b, b, b)\}$.

Binary relations, intuitively

Binary here just means “two”. Relation means we’re talking about a relationship between two items.

Examples:

- “less than”, $a < b$ defines a relationship between two items. The relationship *holds* if and only if a represents a smaller number than b . $5 < 11$, $2021 \not< 1984$.
- “subset”, $A \subseteq B$.
- “is the square of”.
- “is a factor of”.

Binary relations, formally

Definition. A *binary relation*, R , consists of a set, A , called the domain of R , a set, B , called the codomain of R , and a subset of $A \times B$ called the graph of R .

Relation notation

$R : A \rightarrow B$. We say R maps A to B .

(There's a connection with functions, and implications. Hold on for now.)

Kind of like a two-place predicate.

If the relation is from a set A to itself, we say it's a relation “on A ”.

Infix notation is common: $a R b$, meaning (a, b) is in the graph of R .

Binary relation examples

Binary relations are essentially functions that take in two items and return a Boolean.

- superset ($\supset$)?. Yes.
- add (+)? No, binary, but returns a number.
- equals (=)? Yes.
- is divisible by? Yes.
- element of ($\in$)? Yes. Domain and codomain can be different.
- is a perfect square? No, only takes one input. It's a predicate.
- is the perfect square of? Yes.
- greatest common divisor? No, returns a number.
- greatest common divisor is one? Yes, also called “relatively prime” or “co prime”.
- or? Sure, you could see it that way.

Other binary relations and problems

- Is married to?
- Is taller than?
- Is older than?
- Happened before?
- Has met?
- Is faster than?
- Is smarter than?
- Lives with?
- Speaks the same languages as?
- Speaks a same language as?

Defining ordered pairs

We just defined the term “binary relation” in terms of sets: it’s a subset of a product of sets, containing ordered pairs.

But what’s an ordered pair, really?

$(a, b) \neq \{a, b\}$, since $\{a, b\} = \{b, a\}$ but $(a, b) \neq (b, a)$.

Defining ordered pairs

We define the ordered pair (a, b) in terms of sets:

$$(a, b) = \{a, \{a, b\}\}$$

Claim: for any sets X, Y , it cannot be the case that $X \in Y$ and $Y \in X$.

(Why?)

Defining sequences

Once we have ordered pairs, (finite) sequences are easy: just keep pairing.

$$\begin{aligned}(a, b, c) &= ((a, b), c) \\ &= \{(a, b), \{(a, b), c\}\} \\ &= \{\{a, \{a, b\}\}, \{\{a, \{a, b\}\}, c\}\}\end{aligned}$$